# FINAL EXPLANATION: CHEMISTRY GRADE 10

*Student Assessment Document*

| FINAL EXPLANATION: CHEMISTRY GRADE 10 — SUB-STRAND 3.1: ACIDS AND BASES | |
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| **Student Assessment Document** | |
| **Student Name** | ______________________________________ |
| **Class** | ______________________________________ |
| **Date** | ______________________________________ |

## INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION TASK — "The Colour-Changing Drinks of the School Canteen"

Your goal: Write a full, evidence-based explanation that answers the driving question — "What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter for our health, our farms, and our environment?" — using everything you have learned across all 10 lessons.

READ THIS BEFORE YOU BEGIN:

A great Final Explanation is not a list of facts. It tells a story that connects evidence to reasoning. You should explain the phenomenon (the colour-changing cups), defend your explanation with chemical evidence, and show why it matters beyond the classroom. Use the four sections below as a guide. Write in full sentences and paragraphs. Refer back to the eight liquids from the canteen demonstration whenever you can — they are your anchor throughout.

## Section 1 — Defining Acids and Bases: What Is Actually Dissolved in Those Cups?

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| **Explain what acids and bases ARE at the particle level, using the Arrhenius definition. Identify each of the eight canteen liquids (lemon juice, black tea, cow's milk, Stoney Tangawizi, tap water, baking soda solution, liquid soap, and Jik bleach solution) as acidic, basic, or neutral. For each, name the ion that is responsible for its character. Why did the liquids all look similar before the indicator was added, but behave so differently?** | Before the indicator was added, all eight cups looked like ordinary, colourless or lightly coloured liquids — there was nothing you could see that separated a cup of lemon juice from a cup of baking soda solution. Yet looks are deceiving, because what matters is what is dissolved at the particle level, not what the liquid looks like.<br><br>According to the Arrhenius definition, an acid is a substance that dissolves in water and releases hydrogen ions ($H^+$) into the solution. A base is a substance that dissolves in water and releases hydroxide ions ($OH^-$). The lemon juice, Stoney Tangawizi (which contains carbonic and phosphoric acid), black tea, and cow's milk all contain substances that release $H^+$ ions in water — so they are acids. Lemon juice is strongly acidic because citric acid ionises readily, flooding the solution with $H^+$ ions. Stoney Tangawizi is also acidic because dissolved carbon dioxide forms carbonic acid ($CO_2 + H_2O \rightarrow H_2CO_3$), which then releases $H^+$.<br><br>Clean tap water sits in the middle: it releases equal, very tiny amounts of both $H^+$ and $OH^-$ ($K_w = [H^+][OH^-] = 1\times10^{-14}$ at 25 °C), so it is neutral. |

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| | The baking soda solution, liquid soap, and Jik bleach solution all release $OH^-$ ions — they are bases. Jik contains sodium hypochlorite ($NaOCl$), which ionises fully in water and produces a highly alkaline solution. Liquid soap contains sodium or potassium salts of fatty acids that hydrolyse to release $OH^-$. Baking soda ($NaHCO_3$) is a mild base because it partially ionises to release a small amount of $OH^-$.<br><br>The reason all these liquids looked the same before the indicator is that $H^+$ and $OH^-$ ions are invisible — they have no colour and no smell at the concentrations we find in everyday liquids. The universal indicator gave those invisible particles a visible voice: every colour in every cup was caused by the concentration of $H^+$ or $OH^-$ already present before a single drop of indicator touched the liquid. |

| Section 2 — The pH Scale and the Rainbow: Reading the Colours as Numbers | |
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| **Explain how the pH scale works and how it connects to the colours you observed. Use the formula pH = –log[H⁺] to justify why the scale is logarithmic. Assign approximate pH values to each of the eight canteen liquids and match each to its expected indicator colour. Why is it significant that each step on the pH scale represents a tenfold change in $H^+$ concentration — not just a one-unit change?** | The rainbow of colours that appeared in the eight cups was not random — it was a precise map of a mathematical scale called the pH scale, which runs from 0 (most acidic) to 14 (most basic), with 7 representing a perfectly neutral solution at 25 °C.<br><br>The pH of a solution is defined by the formula: $pH = -\log[H^+]$, where $[H^+]$ is the concentration of hydrogen ions in moles per litre (mol/L). The negative sign is used because $[H^+]$ in most solutions is a very small number (e.g., 0.001 mol/L), and taking the negative logarithm converts it into a convenient positive number ($pH = -\log[0.001] = 3$). This means the scale is logarithmic: every single step of 1 pH unit represents a tenfold change in $[H^+]$. So lemon juice at pH 2 has ten times more $H^+$ ions than black tea at pH 3, and one hundred times more than cow's milk at approximately pH 6.5.<br><br>Here is how the eight canteen liquids map onto the scale:<br>– Lemon juice: ~pH 2 → bright red/orange with universal indicator<br>– Stoney Tangawizi: ~pH 3–4 → orange<br>– Black tea (chai): ~pH 5 → orange-yellow<br>– Cow's milk: ~pH 6.5 → yellow-green<br>– Tap water: ~pH 7 → green (the only green cup!)<br>– Baking soda solution: ~pH 8–9 → blue<br>– Liquid soap: ~pH 9–10 → blue-purple<br>– Jik bleach solution: ~pH 11–13 → purple/dark violet<br><br>The fact that the scale is logarithmic is enormously important. It means the difference between lemon juice (pH 2) and Jik bleach (pH 13) is not an 11-unit difference — it is an 11-order-of-magnitude difference, meaning Jik bleach has $10^{11}$ (100 billion) times fewer $H^+$ ions than lemon juice. That is why Jik can destroy pathogens and bleach fabric while lemon juice just flavours your food. The colours in the cups were really telling you about powers of ten — hidden inside what looked like ordinary drinks. |

| Section 3 — Chemical Properties in Action: How Acids and Bases Behave | |
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| **Describe the key chemical properties of acids and bases. Use specific examples from the** | Understanding what acids and bases ARE (Section 1) is only the beginning. To truly explain the phenomenon, you also need to understand what they DO — their chemical properties. |

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| **canteen liquids and from Kenyan life to illustrate each property. Your answer must include: the reaction of acids with metals and carbonates, neutralisation reactions (with a balanced equation), and the behaviour of acids and bases with indicators. Explain how neutralisation is a two-way 'tug-of-war' between $H^+$ and $OH^-$ ions.** | ACIDS react with metals to produce a salt and hydrogen gas. For example, if you added iron nails to lemon juice (citric acid), you would see bubbling as hydrogen gas ($H_2$) is released and an iron salt forms. Farmers in the Rift Valley who store acidic fertiliser improperly sometimes see this corrosion on metal equipment. Acids also react vigorously with carbonates and hydrogen carbonates, releasing carbon dioxide gas. The fizzing you notice when you add lemon juice to baking soda (a bicarbonate) in a Kenyan kitchen is exactly this reaction: the citric acid ($H^+$ ions) reacts with bicarbonate ($HCO_3^-$) to produce $CO_2$ bubbles, water, and a salt.<br><br>BASES feel slippery to the touch (like liquid soap) because $OH^-$ ions react with fatty acids and oils on your skin. Concentrated bases are highly corrosive — this is why Jik bleach carries warnings and why it must never be swallowed.<br><br>NEUTRALISATION is perhaps the most important property. When an acid and a base are mixed in the right proportions, their $H^+$ and $OH^-$ ions cancel each other out in a direct ionic reaction:<br><br>$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$<br><br>This is a 'tug-of-war': the $H^+$ from the acid and the $OH^-$ from the base pull against each other, and if their amounts match exactly, neither wins — the result is pure water (plus a dissolved salt), and the pH reaches 7. If you added baking soda solution to lemon juice drop by drop and watched the indicator, you would see the colour shift gradually from red → orange → yellow → green as the acid is progressively neutralised. A full neutralisation equation for hydrochloric acid and sodium hydroxide reads:<br><br>$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$<br><br>INDICATORS themselves work because they are weak acids or bases whose conjugate forms have different colours — the universal indicator is actually a mixture of several such dyes, which is why it can show so many different colours across the full pH range. Each colour in the canteen cups was the indicator molecules responding to the exact balance of $H^+$ and $OH^-$ already present in the liquid. |

| Section 4 — Why It Matters: Acids, Bases, Health, Farming, and the Environment in Kenya | |
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| **Now zoom out. Connect your chemical understanding to three real-world contexts: (1) human health (e.g., digestion, safe drinking water, use of Jik as a disinfectant), (2) agriculture in Kenya (e.g., soil pH in Kericho tea farms or maize farming in the Rift Valley), and (3) environmental issues (e.g., acid rain, water quality in Lake Victoria or Nairobi rivers). For each context, explain specifically** | The colour-changing cups were not just a classroom trick — they were a window into chemistry that shapes the health of people, the productivity of farms, and the quality of the environment across Kenya every single day.<br><br>(1) HUMAN HEALTH<br>Your stomach produces hydrochloric acid (HCl) at approximately pH 1.5–2, which is similar in acidity to lemon juice. This strong acid is essential for digesting proteins and killing harmful bacteria in food — including pathogens that might be present in street food in Nairobi or Mombasa markets. pH matters here because if the stomach becomes too acidic (pH drops further), the lining is damaged, causing ulcers; if it becomes too basic, digestion fails. Antacid tablets, which contain bases like magnesium hydroxide or sodium bicarbonate, work by neutralising excess stomach acid: $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$. The Jik in the canteen demonstration, with its pH above 11, is too corrosive to drink but is ideal for disinfecting water storage containers and killing cholera bacteria — a vital public health application. Safe drinking water should be close to pH 6.5–8.5; water outside this range can damage pipes, harm gut bacteria, and indicate contamination. |

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| **how the concentration of $H^+$ or $OH^-$ ions affects an outcome that matters. Use the phrase 'pH matters here because…' at least once in each context.** | (2) AGRICULTURE<br>Kenyan tea, grown in Kericho and Nandi Hills, thrives in slightly acidic soil at pH 4.5–5.5. Maize (the main ingredient in ugali), grown widely across the Rift Valley, prefers a pH of 5.8–7.0. pH matters here because soil pH controls which nutrients dissolve and become available to plant roots: in very acidic soils, toxic levels of aluminium and manganese dissolve and poison plants, while in very basic soils, iron and phosphorus lock up and become unavailable. Farmers who apply too much acidic fertiliser (like ammonium sulphate) without liming the soil are unknowingly shifting the pH away from the ideal range, reducing their yields. Lime (calcium hydroxide, $Ca(OH)_2$) is a base that neutralises excess soil acidity — a direct application of the same neutralisation chemistry you saw in the cups.<br><br>(3) ENVIRONMENT<br>Acid rain forms when sulfur dioxide ($SO_2$) and nitrogen oxides ($NO_x$) — released from vehicles and industries in Nairobi and Mombasa — dissolve in rainwater and form sulfuric and nitric acids, dropping the pH of rain from the natural 5.6 to as low as 4.0. pH matters here because aquatic life in rivers and lakes is extremely sensitive to pH: most fish and invertebrates in Lake Victoria tributaries cannot survive below pH 5. Industrial and agricultural runoff also carries acidic or basic waste into rivers, shifting the pH and destroying ecosystems. Monitoring the pH of Nairobi's Athi River or Lake Victoria's inflows is therefore a direct and practical way to measure environmental health — using the exact same chemistry that turned the canteen cups their different colours. |

| **Section 5 — Pulling It All Together: Answering the Driving Question** | |
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| **Write a final, unified paragraph (or two) that directly and completely answers the driving question: 'What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter for our health, our farms, and our environment?' Your answer should weave together the Arrhenius definitions, the pH scale, chemical properties, and real-world significance into one coherent explanation. A reader who had never seen the canteen demonstration should be able to understand the full story from your paragraph alone.** | The eight cups of ordinary liquid that turned a rainbow of colours in the school canteen were revealing something invisible but profoundly important: the concentration and type of charged particles dissolved in each liquid. Liquids like lemon juice and Stoney Tangawizi are acids — they release hydrogen ions ($H^+$) when dissolved in water, and the higher their $H^+$ concentration, the lower their pH and the more intensely red the universal indicator turned. Liquids like liquid soap, baking soda solution, and Jik bleach are bases — they release hydroxide ions ($OH^-$), giving them high pH values and turning the indicator deep blue or purple. Tap water, with equal and minute concentrations of both ions, landed precisely in the middle at pH 7, producing the only green cup. Because the pH scale is logarithmic ($pH = -\log[H^+]$), each colour shift in the cups represented not a small step but a tenfold change in ion concentration — which is why Jik, just a few cups away from lemon juice on the table, is a powerful disinfectant while lemon juice is safe to drink.<br><br>This chemistry is not confined to a classroom demonstration. The same $H^+$ and $OH^-$ ions that coloured those cups control how well your stomach digests ugali and sukuma wiki, whether Kericho tea plants can absorb nutrients from the soil, whether maize crops in the Rift Valley reach their full potential, and whether the fish in Lake Victoria survive industrial runoff. Acids and bases neutralise each other in a precise ionic reaction ($H^+ + OH^- \rightarrow H_2O$), and understanding this balance — measuring it, controlling it, and correcting it when it goes wrong — is one of the most practical skills that chemistry gives us. The colour-changing cups were not magic. They were a pH map of the invisible chemical world that shapes our health, our food, and our environment every single day. |

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| **Criterion** | **Excellent (4)** | **Proficient (3)** | **Developing (1–2)** |

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| Arrhenius Definitions & Ion Identification | Accurately defines both acids (release $H^+$) and bases (release $OH^-$) using the Arrhenius model. Correctly classifies all eight canteen liquids and names the responsible ion for each. Explains clearly why the liquids appeared identical before the indicator was added. | Correctly defines acids and bases and classifies most (6–7) of the eight liquids accurately. Names ions for most liquids. Provides a partial explanation of why the liquids looked similar before testing. | Attempts definitions of acids and bases but contains errors or omissions. Classifies fewer than 6 liquids correctly or confuses $H^+$ and $OH^-$. Little or no explanation of why the liquids appeared the same before the indicator. |
| pH Scale — Logarithmic Reasoning & Colour Mapping | Correctly applies $pH = -\log[H^+]$ and explains the logarithmic nature of the scale with a specific numerical example. Assigns accurate approximate pH values and matching indicator colours to all eight liquids. Clearly explains the significance of the tenfold change per pH unit. | Applies the pH formula correctly and assigns approximate pH values and colours to most liquids. Acknowledges that the scale is logarithmic but explanation of the tenfold significance is general rather than specific. | Shows partial understanding of the pH scale. Formula may be missing or incorrectly applied. Fewer than half the liquids are correctly matched to pH values and colours. The logarithmic nature of the scale is not addressed or is incorrect. |
| Chemical Properties of Acids and Bases | Describes at least three distinct chemical properties (e.g., reaction with metals, reaction with carbonates, neutralisation, effect on indicators). Includes at least one correctly balanced neutralisation equation. Uses specific Kenyan examples to illustrate each property. Explains neutralisation as an ionic $H^+ + OH^- \rightarrow H_2O$ reaction. | Describes two or three chemical properties with mostly correct detail. Includes a neutralisation equation that may have minor balancing errors. Uses at least one Kenyan example. Explains neutralisation in general terms. | Describes one or two properties with significant errors or vague language. Neutralisation equation is absent or incorrect. No Kenyan examples used. Little connection made between properties and the canteen phenomenon. |
| Real-World Application — Health, Agriculture, and Environment | Provides a specific, chemically accurate explanation for all three contexts (health, agriculture, environment) using Kenyan settings (e.g., stomach acid, Kericho tea soil, Lake Victoria). Explicitly connects $H^+/OH^-$ concentration or pH to a measurable outcome in each context. Uses the phrase 'pH matters here because…' or equivalent reasoning in each context. | Addresses all three contexts but one or two lack specific chemical detail or Kenyan grounding. pH is connected to outcomes in most but not all contexts. Language is generally accurate. | Addresses only one or two contexts, or treats all three very superficially. Chemical connection to $H^+/OH^-$ concentration is weak or missing. Kenyan contexts are absent or generic. Does not clearly explain why pH matters. |
| Synthesis — Answering the Driving Question Coherently | The final section (and overall response) tells a complete, coherent story that directly answers the driving question. Weaves Arrhenius definitions, pH scale, chemical properties, and real-world significance together. A reader unfamiliar with the canteen demonstration would understand the full explanation. Writing is clear, precise, and uses correct scientific vocabulary throughout. | The final section addresses the driving question and connects most of the key concepts. The response is mostly coherent but one strand (e.g., real-world significance or pH formula) is underdeveloped. Scientific vocabulary is mostly correct. | The final section attempts to answer the driving question but the response reads as disconnected facts rather than a unified explanation. Key concepts are missing or incorrectly linked. Scientific vocabulary is limited or frequently misused. |